EXPERIMENT -6

RAD- (Rapid Application Developement)

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Project Title: TechStore – E-commerce Web Application

Introduction

The **TechStore E-commerce Application** is a web-based platform that allows users to browse, select, and purchase electronic products such as laptops, audio devices, smartphones, and accessories.

This project follows the **Rapid Application Development (RAD)** model, focusing on quick development, continuous feedback, and iterative design.

Objective

- To design a user-friendly e-commerce interface
- To implement fast and interactive product browsing
- To simplify online purchasing with cart and checkout features
- To demonstrate RAD methodology in UI/UX development

RAD Model Overview

RAD emphasizes:

- Quick prototyping
- User involvement
- Iterative improvements
- Faster delivery

RAD Phases Applied

1. Requirement Planning

- Identified need for an online tech store
- Defined features: product listing, cart, checkout
- Target users: online shoppers

2. User Design

UI was created using Figma with the following screens:

Home Page

- Welcome banner ("Welcome to TechStore")
- Navigation bar (Home, Categories, Cart)
- “Shop Now” button

Category Section

- Laptops
- Audio
- Smartphones
- Cameras
- Wearables
- Accessories

Featured Products Page

- Product cards with:
    - Image
    - Rating
    - Description
    - Price
    - “Add to Cart” button

3. Rapid Construction

- Designed interactive components:
    - Add to cart functionality
    - Quantity adjustment (+ / -)

    - Dynamic price updates
- Created structured layouts for:
    - Cart summary
    - Checkout form

Cutover (Final Stage)

- Final UI screens completed
- Ready for deployment and testing
- User-friendly navigation ensured

System Modules

User Module

- Basic navigation (no login shown but extendable)

Product Module

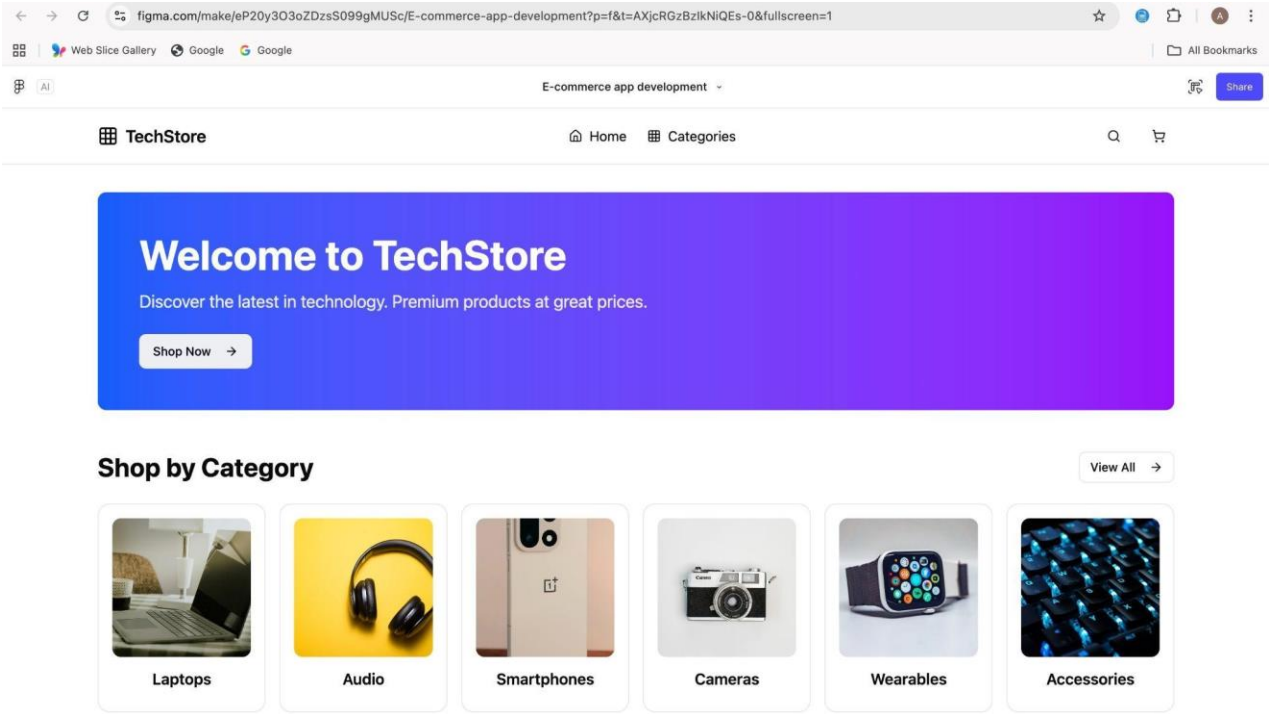
- Displays all available products
- Shows details like price, rating, description

Cart Module

- Add/remove items
- Increase/decrease quantity
- Displays subtotal, tax, shipping

Checkout Module

- Collects user details:
    - Name
    - Email
    - Phone
    - Address
- Shows final order summary
- “Place Order” option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Home
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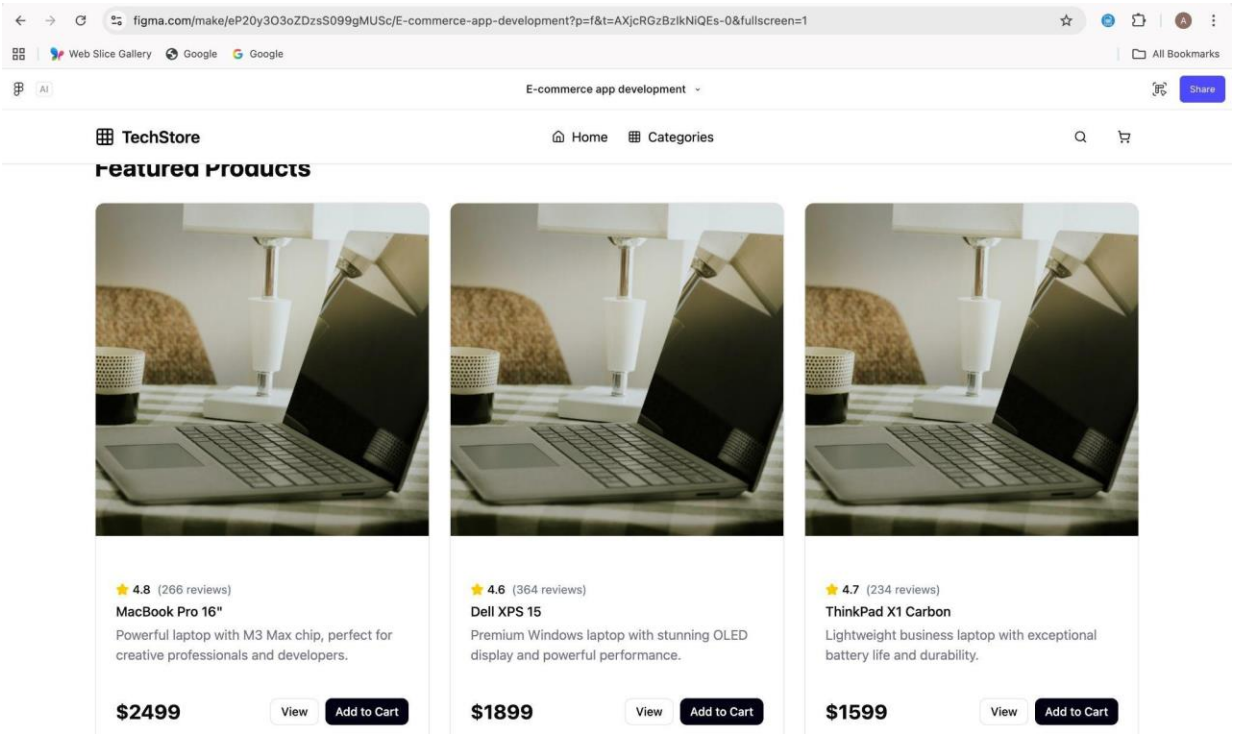
Discover the latest in technology. Premium products at great prices.
Shop Now
Shop by Category
View All
Laptops
Audio
Smartphones
Cameras
Wearables
Accessories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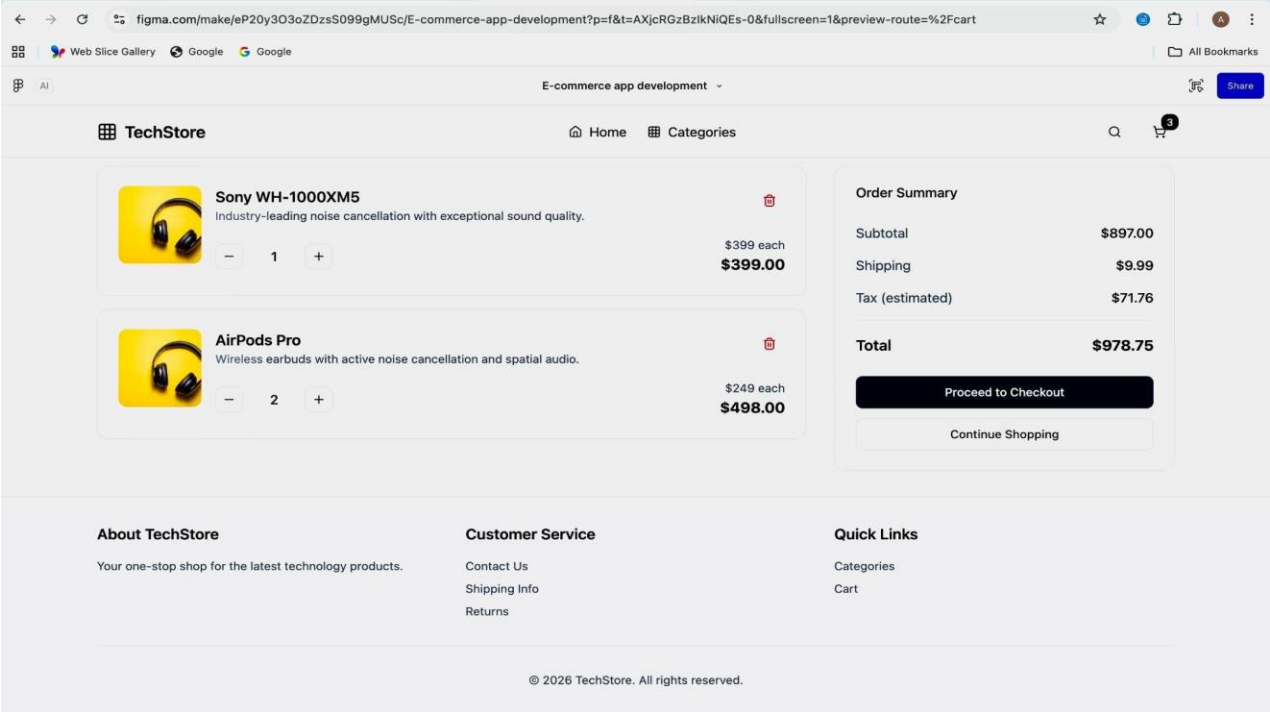
Web Slice Gallery
Google
Google
All Bookmarks
E-commerce app development
Share
TechStore
Home
Categories
Featured Products
4.8 (266 reviews)
MacBook Pro 16"
Powerful laptop with M3 Max chip, perfect for creative professionals and developers.
$2499
View
Add to Cart
4.6 (364 reviews)
Dell XPS 15
Premium Windows laptop with stunning OLED display and powerful performance.
$1899
View
Add to Cart
4.7 (234 reviews)
ThinkPad X1 Carbon
Lightweight business laptop with exceptional battery life and durability.
$1599
View
Add to Cart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Home
Categories
3
Sony WH-1000XM5
Industry-leading noise cancellation with exceptional sound quality.
1
$399 each
$399.00
AirPods Pro
Wireless earbuds with active noise cancellation and spatial audio.
2
$249 each
$498.00
Order Summary
Subtotal $897.00
Shipping $9.99
Tax (estimated) $71.76
Total $978.75
Proceed to Checkout
Continue Shopping
About TechStore
Your one-stop shop for the latest technology products.
Customer Service
Contact Us
Shipping Info
Returns
Quick Links
Categories
Cart
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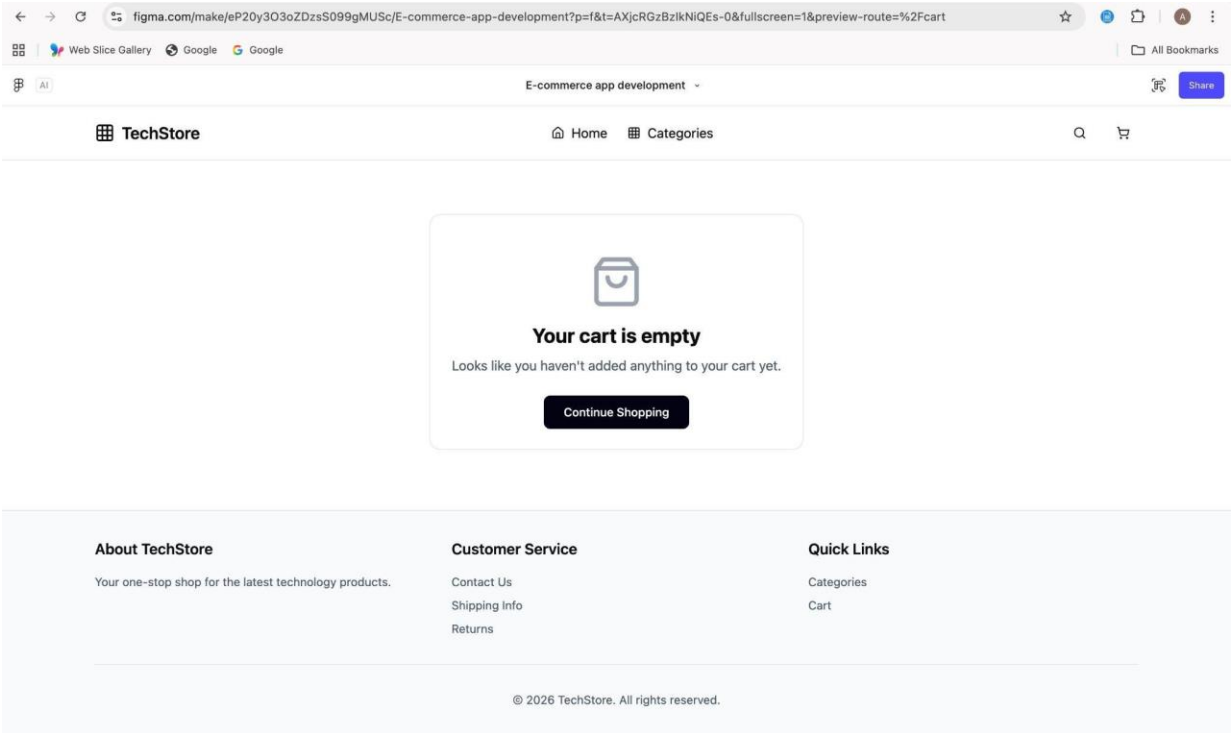
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Web Slice Gallery
Google
Google
All Bookmarks
AI
E-commerce app development
Share
TechStore
Home
Categories
Your cart is empty
Looks like you haven't added anything to your cart yet.
Continue Shopping
About TechStore
Your one-stop shop for the latest technology products.
Customer Service
Contact Us
Shipping Info
Returns
Quick Links
Categories
Cart
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Workflow of the System

1. User visits homepage
2. Browses categories
3. Selects a product
4. Adds product to cart
5. Reviews cart details
6. Proceeds to checkout
7. Enters personal & shipping details
8. Places order

UI Design Highlights (From Your Screens)

- Clean and modern layout
- Gradient banner for visual appeal
- Card-based product design
- Simple navigation bar
- Responsive grid system
- Clear call-to-action buttons

Tools & Technologies

- **Design Tool:** Figma
- **Frontend:** HTML, CSS, JavaScript (conceptual)
- **Backend (optional):** Node.js / Firebase
- **Database:** MongoDB / Firestore

Advantages of the System

- Easy product browsing
- Smooth user experience
- Fast checkout process
- Scalable for future features

Future Enhancements

- User login & authentication
- Payment gateway integration
- Order tracking system
- Wishlist feature
- AI-based recommendations

Conclusion

The **TechStore E-commerce application** successfully demonstrates the **RAD model**, allowing rapid UI development with continuous improvements. The system is efficient, scalable, and user-friendly, making it suitable for real-world implementation.